Assignment 2: Project Updates (3 total)

1. Progress Summary

Based on our project proposal document, we achieved the goals we had outlined for Project Update 1. In terms of benchmark development, we created a base JSON world file. This file will be used as a guide while developing the 20 different benchmark scenarios/tasks. We also wrote the first end-to-end benchmark scenario related to basic triage. In terms of agent development, we wrote the World Tools for the agent to interact with the map and implemented the Validator component to allow the Python script to check the LLM's logic and actions.

2. Current Status

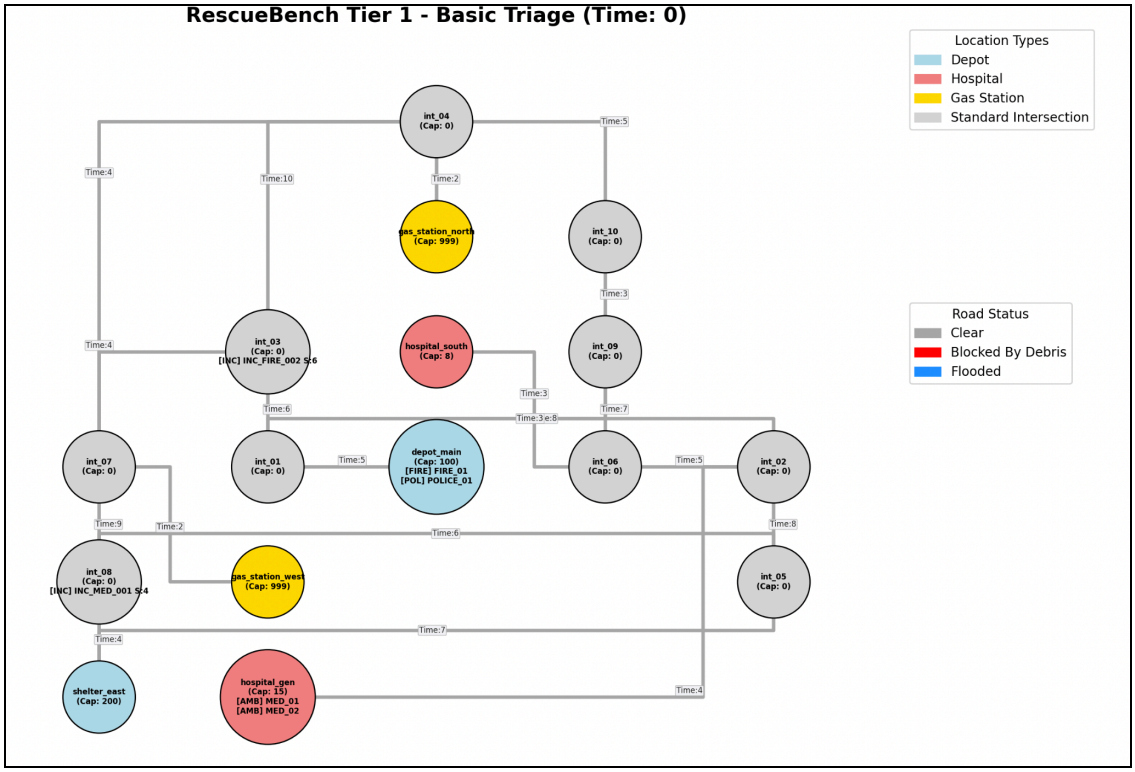
Benchmark: Base JSON world created, first benchmark task designed and documented (20%)

Agent: WorldState (JSON loader, graph builder, Dijkstra's shortest path), WorldTool (6 query/action functions: get_map_state, get_vehicles, get_incidents, get_shortest_path, dispatch_vehicle, report_status), and ValidatorTool (7-check pre-execution safety layer) are all complete.

Evaluation: No tests run yet. The scoring formulas for Priority-Weighted Resolution Score and Hallucination/Violation Penalty are integrated into the agent's final output, but have not been validated against a live API call.

3. Challenges and Solutions

1. After creating the base JSON world and first benchmark scenario, the city graph was a bit complicated to visualize. A python script was written specifically to visualize any benchmark scenario's city graph and output a clear, labeled and accurate image using the NetworkX library.



2. WorldState was tricky to design since it tracks so many things at once (graph, blocked edges, vehicles, incidents). This was solved by making the Validator read-only, so only WorldTool could ever mutate state.

4. Next Steps

1. Complete the first 5 tiered benchmark problems
2. Run a naive baseline attempt
3. Reevaluate prompting strategies
4. Confirm and finalize evaluation metrics

5. Questions and Help Needed

We may require assistance on confirming the evaluation metrics after we test our agent end-to-end on the first 5 benchmark problems.

6. Individual Contributions

A brief statement from each team member: - One sentence per person describing what that individual did this week.

- **Akshat Bhat [akshatb5]:**

  Successfully designed the base_city_world.json master template, establishing a complex 16-node city graph complete with heterogeneous vehicle capacities, weight-restricted edges, and dynamic environmental triggers.

- **Trusha Talati [ttalati2]:**

  Generated the first official Level 1 benchmark scenario (tier1_basic_triage.json), utilizing the master template, by stripping out the dynamic mid-mission updates to focus the agent purely on basic triage and straightforward single-unit routing.

- **Akhil Gogineni [vg23]:**

  Implemented the full agent architecture in a single Python module, including the WorldState simulation engine (JSON loader + Dijkstra's), WorldTool (6 query/action functions), ValidatorTool (7-check deterministic safety layer), the Claude API tool schema definitions, and the ReAct orchestration loop with self-correction feedback.

- **Nicolas Yepes [nyepes2]:**

  Worked on ValidationTool and integrating with world schema / city graph. Constraints the LLM to certain actions so that it can generate valid answers, Thoroughly tested for accuracy.